Ab initio calculations of two-neutrino and neutrinoless double-β decay of ^{48}Ca and related Gamow–Teller strength distributions

Zhen Li   Lotta Jokiniemi   Achim Schwenk

Technische Universität Darmstadt
ExtreMe Matter Institute EMMI, GSI
Max-Planck-Institut für Kernphysik

Journal Club Presenter: Jiangming Yao

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Outline

1. Motivation: why $2\nu\beta\beta$ and $0\nu\beta\beta$ matrix elements matter
2. Theoretical framework: VS-IMSRG with chiral interactions and weak currents
3. Main result I: enlarged valence space resolves $2\nu\beta\beta$ strength in ^{48}Ca
4. Main result II: impact on $0\nu\beta\beta$ nuclear matrix elements
5. Conclusions and perspectives for heavier candidates

Physics motivation

Double-β decay

- $2\nu\beta\beta$: rare Standard-Model process with measured half-lives.
- $0\nu\beta\beta$: lepton-number-violating process; observation would imply Majorana neutrinos.
- Interpreting a signal requires reliable nuclear matrix elements (NMEs).

Why ^{48}Ca?

- Lightest double-β emitter and a benchmark for ab initio methods.
- Allows direct connection between $2\nu\beta\beta$, $0\nu\beta\beta$, and GT strength in ^{48}Sc.

Central issue

Phenomenological NME predictions differ substantially. Ab initio calculations provide a systematically improvable route, but their accuracy depends on correlations captured by the many-body truncation and valence space.

Abstract-level message of the manuscript

Main finding

The usual *pf*-shell valence space underestimates the experimental $2\nu\beta\beta$ NME of ^{48}Ca, while the enlarged $d_{3/2}pf$ space gives good agreement without phenomenological adjustment.

- Improvement is traced to a better description of GT strength distributions in ^{48}Sc.
- Chiral two-body currents are included for $2\nu\beta\beta$.
- The enlarged $d_{3/2}pf$ valence space increases $0\nu\beta\beta$ NMEs by about a factor of two.
- Extended valence spaces and GT data are essential diagnostics for heavier candidates.

Theoretical framework: observables

$2\nu\beta\beta$ half-life

$$\left(T_{1/2}^{2\nu}\right)^{-1} = G_{2\nu} g_A^4 \left(m_e M^{2\nu}\right)^2 .$$

$$M^{2\nu} = \sum_k \frac{\langle 0_f^+ || GT^- || 1_k^+ \rangle \langle 1_k^+ || GT^- || 0_i^+ \rangle}{E_k - (E_{\mathrm{gs}}^i + E_{\mathrm{gs}}^f)/2} .$$

$0\nu\beta\beta$ half-life

$$\left(T_{1/2}^{0\nu}\right)^{-1} = G_{0\nu} g_A^4 \left(M^{0\nu} \frac{m_{\beta\beta}}{m_e}\right)^2 .$$

$$M^{0\nu} = M_{\mathrm{LR}}^{0\nu} + M_{\mathrm{SR}}^{0\nu} .$$

- Long-range light-neutrino exchange.
- Short-range contact term from chiral EFT matching.

Theoretical framework: VS-IMSRG calculation

Inputs

- Chiral NN+3N interactions:
  - 1.8/2.0 (EM)
  - ΔN^{2}LO$_{\mathrm{GO}}$ (394)
- Weak currents from chiral EFT.
- Leading two-body currents included up to N^{2}LO for $2\nu\beta\beta$.

Many-body method

1. Hartree–Fock basis with $e_{\mathrm{max}} = 12$ baseline.
2. VS-IMSRG(2) decouples a chosen valence space.
3. Diagonalize the evolved Hamiltonian using shell-model methods.
4. Evolve GT and decay operators consistently.

Valence spaces: the key comparison

pf shell

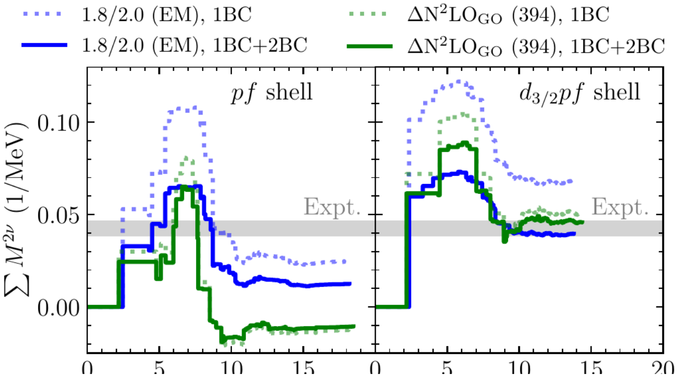
- Standard choice on top of a ^{48}Ca[40] core.
- Earlier VS-IMSRG ^{48}Ca double-β calculations used this space.
- Missing correlations can appear as induced higher-body operators in VS-IMSRG(2).

Enlarged $d_{3/2}pf$ shell

- Adds the $0d_{3/2}$ orbital on top of a ^{32}S core.
- Removes spurious c.m. motion with a Lawson term.
- Uses a $4\hbar\omega$ truncation for $2\nu\beta\beta$ running sums.

Question: can the enlarged space recover the GT correlations needed for $2\nu\beta\beta$ and $0\nu\beta\beta$?

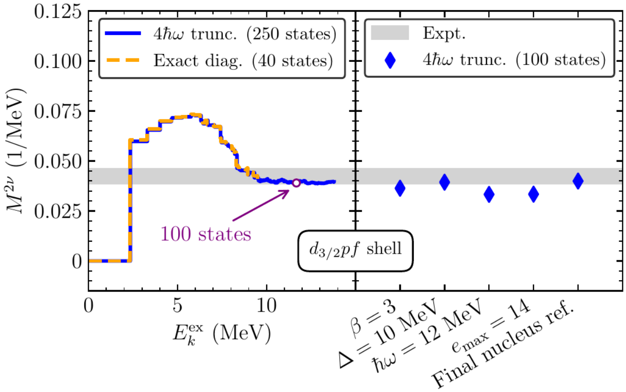
Main result I: $2\nu\beta\beta$ running sum



Observation

- In the *pf* shell, VS-IMSRG underestimates the experimental $2\nu\beta\beta$ NME.
- In the $d_{3/2}pf$ space, the running sum reaches the experimental band for both interactions when 2BCs are included.
- The enlarged space enhances the low-energy positive contribution and reduces the high-energy cancellation.

Robustness of the enlarged-space result



Tests performed

- $4\hbar\omega$ truncation vs exact diagonalization.
- Variation of Lawson parameter β.
- Variation of denominator shift Δ.
- Variation of $\hbar\omega$, e_{max}, and reference nucleus.

Conclusion

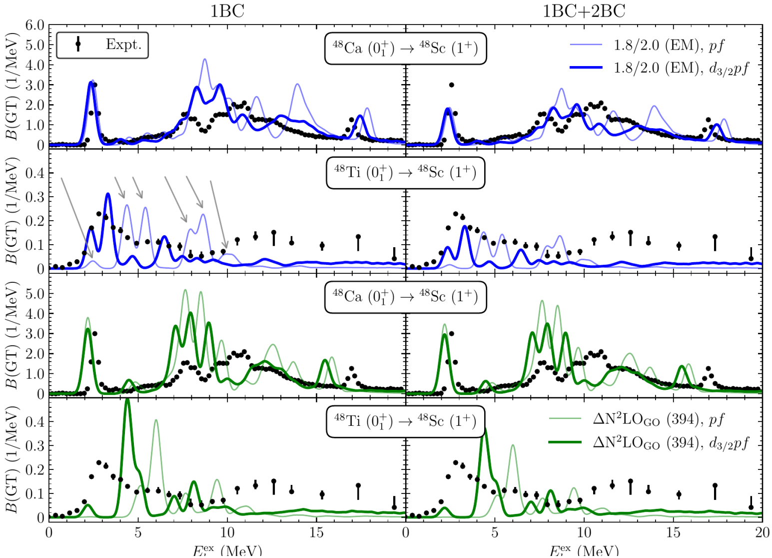
The improvement from the $d_{3/2}pf$ space is stable against these calculational choices.

Quantitative $2\nu\beta\beta$ NMEs

		pf shell		$d_{3/2}pf$ shell	
Current	Interaction	1.8/2.0 (EM)	ΔN^2LO_{GO}	1.8/2.0 (EM)	ΔN^2LO_{GO}
1BC	$\lvert M^{2\nu}\rvert$	0.0246	0.0124	0.0681	0.0504
1BC+2BC	$\lvert M^{2\nu}\rvert$	0.0125	0.0106	0.0395	0.0458

- Units: MeV^{-1}.
- The experimental value is reproduced only after enlarging the valence space.
- Two-body currents reduce the 1.8/2.0 (EM) result substantially and bring the two interactions closer in the $d_{3/2}pf$ space.

Why does the enlarged space help? GT strengths



Mechanism

- Compare calculated GT strengths with ^{48}Ca(p,n) and ^{48}Ti(n,p) data.
- The $d_{3/2}pf$ space broadens and shifts key peaks toward experiment.
- Strength is enhanced below $\sim$ 7 MeV and weakened in the region that drives cancellation.

Impact of two-body currents (2BC) on the Sums

Table: Ratios of $\sum_f B(\mathrm{GT}^-)$ in the β decay of $^{48}\mathrm{Ca}$.

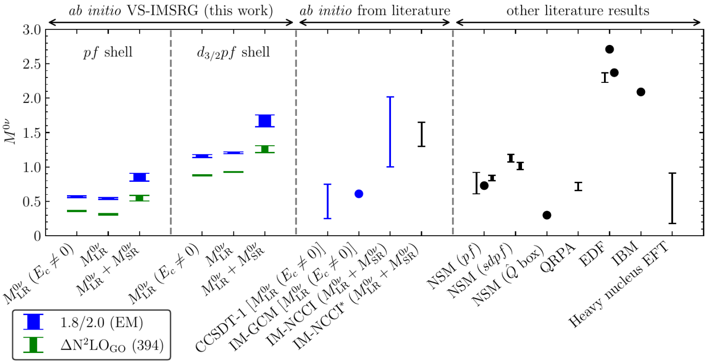
Interaction	imsrgGT1B/bareGT1B		imsrgGT1B2B/bareGT1B		imsrgGT1B2B/imsrgGT1B	
	pf	$d_{3/2}pf$ $(4\hbar\omega)$	pf	$d_{3/2}pf$ $(4\hbar\omega)$	pf	$d_{3/2}pf$ $(4\hbar\omega)$
1.8/2.0 (EM)	0.854	0.783	0.561	0.516	0.657	0.659

Table: Square roots of the ratios of $\sum_f B(\mathrm{GT}^-)$ in the β decay of $^{48}\mathrm{Ca}$.

Interaction	$(\mathrm{imsrgGT1B/bareGT1B})^{1/2}$		$(\mathrm{imsrgGT1B2B/bareGT1B})^{1/2}$		$(\mathrm{imsrgGT1B2B/imsrgGT1B})^{1/2}$	
	pf	$d_{3/2}pf$ $(4\hbar\omega)$	pf	$d_{3/2}pf$ $(4\hbar\omega)$	pf	$d_{3/2}pf$ $(4\hbar\omega)$
1.8/2.0 (EM)	0.924	0.885	0.749	0.718	0.811	0.812

Zheng Li, private communication, 2026

Main result II: impact on $0\nu\beta\beta$ NMEs



Key result

- The $d_{3/2}pf$ space roughly doubles $M^{0\nu}$ compared with the pf-shell calculation.
- The combined LR+SR NME aligns more closely with IM-NCCI results.
- Since half-life scales as $1/|M^{0\nu}|^2$, a factor-two NME increase implies a factor-four shorter half-life for fixed $m_{\beta\beta}$.

Quantitative $0\nu\beta\beta$ NMEs

NME	*pf* shell		$d_{3/2}pf$ shell	
	1.8/2.0 (EM)	ΔN^{2}LO$_{\rm GO}$	1.8/2.0 (EM)	ΔN^{2}LO$_{\rm GO}$
$M^{0\nu}_{\rm LR}(E_c \neq 0)$	0.557–0.579	0.351–0.366	1.135–1.180	0.872–0.886
$M^{0\nu}_{\rm LR}$	0.528–0.556	0.302–0.322	1.192–1.219	0.924–0.929
$M^{0\nu}_{\rm LR} + M^{0\nu}_{\rm SR}$	0.796–0.907	0.507–0.589	1.582–1.756	1.207–1.310

- Ranges probe basis/reference choices and short-range coupling uncertainty.
- The short-range contribution is numerically important.
- Extended valence-space correlations matter for both $2\nu\beta\beta$ and $0\nu\beta\beta$.

Interpretation and implications

Nuclear-structure lesson

- $2\nu\beta\beta$ is sensitive to intermediate 1^+ states and GT fragmentation.
- The *pf* shell misses correlations that are recovered partly by adding $0d_{3/2}$.
- GT strength distributions provide a powerful diagnostic for NME quality.

For heavier candidates

- Similar studies are needed for ^{76}Ge, ^{100}Mo, ^{130}Te, and ^{136}Xe.
- Extended valence spaces can shift predicted $0\nu\beta\beta$ NMEs significantly.
- Higher-body IMSRG operators are a natural next step.

Conclusions

1. Ab initio VS-IMSRG calculations of ^{48}Ca double-β decay were performed with chiral NN+3N interactions and consistent weak currents.
2. The standard *pf* shell underestimates the $2\nu\beta\beta$ NME; the enlarged $d_{3/2}pf$ valence space gives good agreement with experiment.
3. The improvement is explained by a better description of GT strengths in ^{48}Sc.
4. The enlarged valence space increases $0\nu\beta\beta$ NMEs by about a factor of two.
5. Future progress requires extended valence spaces, GT-strength benchmarks, and higher-body operators in IMSRG.

Suggested discussion questions

- Which correlations are most responsible for the improvement from *pf* to $d_{3/2}pf$?
- Can GT charge-exchange data be used systematically to constrain or validate ab initio $0\nu\beta\beta$ NMEs?
- How large are the missing IMSRG(3) and induced three-body-operator effects?
- What is the most efficient path to extended-space calculations for ^{76}Ge, ^{100}Mo, ^{130}Te, and ^{136}Xe?